Data Science and Engineering (2025) 10:495–514
https://doi.org/10.1007/s41019-025-00288-9

RESEARCH PAPERS

An Exploratory Study on Information Cocoon in Recommender Systems

Lin Zhang[1] · Yahong Lian[1] · Haixia Wu[1] · Chunyao Song[1] · Xiaojie Yuan[1]

Received: 30 November 2024 / Revised: 21 February 2025 / Accepted: 16 March 2025 / Published online: 19 May 2025
© The Author(s) 2025

Abstract
In recent years, while algorithm-driven recommendation applications have seen widespread use, their negative impacts have also increasingly raised concerns. To gain a more comprehensive understanding of the impact of different recommendation algorithms, we explored the phenomenon of information cocoons, where users are enveloped by homogenized recommended content, in different algorithm-driven recommender systems. We simulated long-term interactions between users and various algorithm-driven recommender systems, trying to recreate multi-stage recommendation scenarios under the influence of complex factors, and explored whether and to what extent users would fall into information cocoons while analyzing the underlying reasons from the perspective of algorithms. We conducted simulation experiments on two real-world recommendation datasets from different fields. The results show that information cocoons is prevalent across different algorithm-driven recommender systems, and the extent of its occurrence varies. Diversity-oriented recommendations can help alleviate information cocoons but are limited in effectiveness. The ability of diversity-aware re-ranking frameworks to alleviate information cocoons is influenced by the basic recommendation models. Not only considering the diversity of the current recommendation list but also the similarity between items and users' historical consumption content, we proposed a simple and lightweight re-ranking framework called ICMF. Compared to other re-ranking methods, ICMF avoids an average of 12.48% of users encountering homogenized recommended content.

Keywords Information cocoon · Recommender system · Simulation · Recommendation homogenization

1 Introduction

In the era of big data with information overload, algorithm-driven recommenders serve as efficient information filtering tools that help filter out potentially interesting content for users. Various recommendation applications have become the main channels for people to access information and consume daily. However, recommendation algorithms inevitably bring some negative impacts, and the effect of information cocoons has attracted increasing attention [31, 52, 55]. In the long-term interaction with the recommender system, users may encounter increasingly narrow content and gradually be enveloped by homogenized content [49]. This not only leads to a suboptimal user experience, for example, user fatigue [32], but also hinders the spread of content on the platform [34]. Moreover, being restricted to limited information for a prolonged period, users may be trapped in their own information cocoons, facing risks such as cognitive rigidity or polarization on social media [79] or politics. Research also indicates that most users in the system hold a negative

Lin Zhang and Yahong Lian: Co-first author.

✉ Chunyao Song
chunyao.song@nankai.edu.cn

Lin Zhang
zhanglin@dbis.nankai.edu.cn

Yahong Lian
yahong.lian@mail.nankai.edu.cn

Haixia Wu
wuhaixia@dbis.nankai.edu.cn

Xiaojie Yuan
yuanxj@nankai.edu.cn

[1] College of Computer Science, TJ Key Lab of NDST, DISSec, TMCC, TBI Center, Nankai University, Tianjin 300350, China

attitude towards the information cocoon effect and hope to suppress it [36].

The similar concepts of "filter bubbles" and "echo chambers" have been proposed online. "Filter bubble" is a broad concept that describes the multifaceted adverse effects of personalized recommendations by affecting information dissemination, including homogenization of recommended content [2, 22, 40], information dissemination bias [37, 51], ideological segregation among users [15, 44], and polarization of user consciousness and social networks [10, 19, 60]. "Echo chamber" is modeled as a homogenized community with similar interests or opinions [11, 20, 57], which can permeate information with similar ideologies while excluding opposing views [13, 45, 46], leading to the repeated reinforcement of similar beliefs within a closed community, and even causing social polarization [14, 53, 56]. To differentiate from existing concepts, in this work, we refer to the concept of "information cocoons", studying the longitudinal impact of different recommendation algorithms on the homogenization of user interaction content.

Personalized recommendation algorithms utilize user features and historical interaction data to mine user interests and provide precise recommendations. However, excessive fitting to the user's historical interaction data may lead to the repetitive appearance of similar content. And this effect can be further exacerbated under the influence of a long-term feedback loop [28, 42, 59, 64], i.e., the continuous mutual influence between users and the recommender system. Figure 1 illustrates the feedback loop and the potential changes in user interests in the recommender system. Users give feedback on the content presented by the recommendation platform, and their feedback data is then used to update the recommendation model, forming a feedback loop. In this process, users may be exposed to increasingly homogeneous recommended content, leading to a narrowing of their consumption interests.

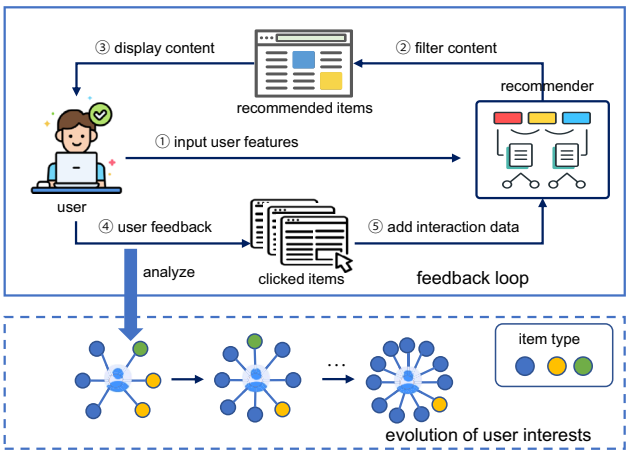


Fig. 1 Feedback loop in the recommender system

As the information cocoon effect becomes increasingly prominent in various fields, it has attracted growing research attention [22, 31, 61]. Some studies point out that recommendation technology is the main cause of information cocoons. Piao et al. [52] revealed through the construction of a mechanistic model that the adaptive process between users and algorithms leads to the emergence of information cocoons. Zhang et al. [84] built a two-way game model between users and recommendation platforms, discovering that recommendation algorithms are the root cause of information cocoons. However, there is still a research gap in understanding how different types of recommendation algorithms affect the formation of information cocoons. With the advancement of deep learning techniques, various recommendation algorithms have emerged [4, 27, 76], which learn user behavior patterns and preferences based on different internal mechanisms. Exploring whether and to what extent different recommendation algorithms may affect the formation of information cocoons is of significant importance in gaining a comprehensive understanding of the algorithmic impact.

A feasible approach to exploring the long-term evolutionary patterns between users and recommender systems under different conditions is to simulate user-system interactions [37, 42], and collect and analyze user interaction data under various conditions. Some work simulates user decision-making based on numerical simulation [3], causal inference [7], interest matching [22], etc. However, these methods do not take into account the uncertainty of user decisions, which can be easily influenced by factors such as sudden interest or environmental background. Besides, the number of candidate items in the real world is vast, when faced with massive candidate items, recommender systems typically perform multi-stage tasks such as recall, ranking, and re-ranking. The uncertainty of user behaviors and the coarse-grained item matching may affect the diversity of recommended content, thus needing consideration in the simulation framework. Moreover, in long-term interaction scenarios, the recommendation model needs to be updated in real-time based on new interaction data. Therefore, to more accurately reproduce the behavioral dynamics of algorithm-driven recommender systems, we propose a versatile simulation framework for long-term interactions between users and recommender systems. It is worth noting that real-world recommendation scenarios are complex and variable, making covering all factors affecting recommendation results impossible. We consider the key factors influencing recommendation diversity and simplify other details.

In this study, we focus on whether the content recommended to users by recommendation algorithms becomes increasingly limited and whether users gradually fall into a homogenous information cocoon during their long-term interactions with the recommender

system. From the perspective of algorithmic causes, we explore whether and to what extent different types of recommendation algorithms lead to information cocoons, and whether existing diversity-oriented recommendation algorithms can mitigate information cocoons and their effectiveness. We aim to raise awareness of the issue of information cocoons, analyze and identify the key algorithmic causes leading to information cocoons, gain a more comprehensive understanding of algorithmic impact, and clarify the limitations and potential improvements of current methods in addressing the information cocoon problem.

To study this issue, we proposed a novel simulation method for user-recommender system interactions and conducted multiple rounds of interaction simulations. Due to the lack of interpretability of the internal hyperparameter mechanism of algorithm design [22], we focus on the outer interaction between users and the algorithm-driven recommender system. Based on this, we investigated the biases caused by different recommendation algorithms and further explored the reasons behind the resulting variations. The experimental results show that users fall into varying degrees of information cocoons under the influence of different recommendation algorithms. In addition, we also demonstrated that although existing diversity-oriented recommendation methods can alleviate information cocoons to a certain extent, there is still room for improvement. In this regard, we also proposed a lightweight Information Cocoon Mitigation Framework (ICMF) to alleviate the homogenization of recommended content. The main contributions of this work can be summarized as follows:

- We propose a simulation framework for modeling the long-term interactions between users and recommender systems, which considers important factors affecting the diversity of recommended content in the real world, and studies the phenomenon of user-centered information cocoons under the influence of different types of recommendation algorithms.
- We conduct experiments on two recommendation datasets from different domains and investigate three important issues: (1) To what extent do different recommendation algorithms cause information cocoons? (2) What are the potential reasons for recommendation algorithms causing information cocoons? (3) What is the relationship between diversity-oriented recommendation and the formation of information cocoons?
- We propose a lightweight re-ranking framework called ICMF (Information Cocoon Mitigation Framework) to mitigate information cocoons. The experimental results demonstrate that, compared to existing re-ranking frameworks, ICMF not only effectively reduces the number of users receiving homogeneous content but also slows down the pace of homogeneity of recommended content.

2 Related Work

2.1 Information Cocoon in Recommendation

Recently, with the wide application of recommender systems, issues such as information cocoons, filter bubbles, and poisoning threats have gradually gained attention [2, 52, 81, 82]. In this work, we focus on the longitudinal impact of recommender systems on users, the information cocoon, where users receive increasingly homogeneous information. Currently, research on information cocoons is still limited, and there are some studies related to the homogenization of recommended content within the context of filter bubbles and echo chambers. Existing methods for detecting information cocoons primarily rely on data quantification [48, 50]. Quantitative analysis studies have demonstrated the prevalence of the information cocoon phenomenon across various recommendation scenarios, including music recommendation [1, 38], news recommendation [43], e-commerce recommendation [18], and short video recommendation [31, 61]. Lunardi et al. [40] developed a news recommendation prototype to assess the impact of a group of recommendation algorithms on recommendation diversity; however, the recommendation algorithms involved were simple. SSLE [22] deployed robots in three news recommendation applications to simulate user-system interaction, revealing a narrowing of the news topics users are exposed to over time. Wang et al. [69] verified the existence of information cocoons using large language model-based agents. Additionally, Sheng et al. [58] analyzed the formation and avoidance methods of information cocoons based on user interviews. These studies, using different strategies, provide feasible approaches to identify information cocoons.

Some research has also explored the causes of information cocoon formation. Piao et al. [52] revealed through constructing an adaptive dynamic model that the similarity between user preferences and recommended content, as well as positive feedback, are important causes of information cocoon formation. Zhang et al. [84], based on a game-theoretical model of the co-evolution between users and recommendation platforms, discovered that the mechanism of recommendation algorithms determines the formation of information cocoons. Currently, various types of recommendation algorithms have been developed and applied to varying extents. The impact of different types of recommendation algorithms on information cocoons may be inconsistent, as they are based on different recommendation mechanisms. However, the influence of different types of

recommendation algorithms on the formation of information cocoons has not yet been explored.

2.2 Simulation of User-Recommender System Interaction

Simulating the long-term interaction between users and the recommender system can help discover the underlying evolutionary patterns. Aridor et al. [3] studied the theoretical model of user decision-making in the context of recommender systems through numerical simulations. Chaney et al. [7] applied causal inference techniques to model multiple interactions between users and the recommender system. However, the aforementioned work does not model the uncertain behaviors of users. Considering this, some studies simulate users' uncertain behaviors by generating random numbers. Mansoury et al. [42] predicted the user's click probabilities and ratings based on the item's ranking in the recommendation list. But this is too idealistic compared to the actual user decision-making situation. SIREN [5] determines user click probabilities based on the distance between users and articles in a two-dimensional feature space and article prominence. In the context of political news recommendation, Liu et al. [37] determined user click probabilities by matching the user's political positions and news topics. With the emergence of large model technologies, Wang et al. [69] used large language models to simulate user behaviors. However, existing simulation studies do not replicate the real-world multi-stage recommendation scenarios, and also overlook the dynamic updates of recommendation models.

2.3 Mitigating the Information Cocoon

As the issue of information cocoons gradually gains attention, some research has emerged to help alleviate this problem [55, 58]. Gao et al. [17] proposed a VAE-based recommender system designed to enhance recommendation diversity along targeted dimensions while preserving relevance. Sun et al. [63] developed a cross-domain matrix factorization recommendation model based on adaptive diversity regularization. Luo et al. [41] proposed a twin-structure framework applied to GNNs for mitigating the bias of cross-links between node clusters, which helps alleviate information cocoons and maintain the connectivity of the graph. However, these models do not directly address the issue of information cocoons and are limited to specific application scenarios. Differently from existing works, Wang et al. [70] proposed a recommender system in which users can actively alleviate content homogeneity, but the applicable models are limited. Santos [55] proposed a new theoretical model to capture the information dynamics driven by algorithmic recommendation and explored measures to alleviate information cocoons based on this model. The results indicate that users' active exploration and negative feedback can enhance the diversity of recommended content. This motivates further research in these areas.

2.4 Diversity-Oriented Recommendation

This type of recommendation method naturally benefits from alleviating the information cocoon, and therefore, we discuss some related studies in this regard. From the technical perspective, diversity-oriented recommendation can be divided into end-to-end methods [33, 35, 39, 62, 80, 85] and post-processing methods [6, 8, 16, 86]. End-to-end methods consider both accuracy and diversity during the model training and inference process to enhance recommendation diversity. EDUA [35] constructs a bilateral branch network based on metric learning for dual-objective optimization. DivMF [30] is an aggregately diversity-oriented recommendation model based on matrix factorization, generating diverse recommendations through coverage penalty and skewness penalty constraints. Moreover, some diversity-oriented recommendation models are based on graph neural networks, such as GraphDR [77] and DGRec [78].

On the other hand, post-processing methods optimize recommendation diversity during the ranking or re-ranking stage and can be flexibly combined with other recommendation models. DDRN [72] is a diversity-aware deep ranking network applied at the ranking stage. MMR [6] is a maximal marginal relevance re-ranking model, used to balance recommendation relevance and diversity. Chen et al. [8] transformed the maximization of the diversity problem into a determinantal point process and solved it based on a greedy strategy. Hirata et al. [25] proposed the problem of diversity-aware maximum inner product search and provided an approximate solution algorithm named IP-Greedy. Coppolillo et al. [12] proposed a probabilistic user behavior model to optimize diversity while considering relevance and user behavior. Wang et al. [68] studied the problem of maximizing sequential diversity and provided theoretical analysis and treatment. Furthermore, Huang et al. [26] proposed the sliding spectrum decomposition model in the content feed recommendation scenario, capturing users' diversity needs while browsing long item sequences. Most existing re-ranking methods aim to improve the diversity of recommendation lists but overlook the similarity between recommended items and users' historical interaction items, and focus on short-term results rather than long-term results. In this work, from a long-term interaction perspective, we studied the ability of several representative diversity-oriented recommendation models to mitigate information cocoons, and proposed a re-ranking method considering user interaction history to improve recommendation diversity.

2.5 Diversity Metric

This work focuses on the changes in individual-level diversity over time, from which the formation process of information cocoons can be analyzed. Individual-level diversity can be divided into two subcategories. The first type includes distance-based metrics such as cosine diversity distance [9], Jaccard diversity distance [65], etc. The second type is coverage-based metrics like prediction coverage [54], catalog coverage [85], subtopic coverage [24], and others. Lunardi et al. [40] quantified the homogenization of recommended content by calculating the average Jaccard coefficient between pairs of recommended items. Vrijenhoek et al. [66] proposed the distance-based diversity metric RADio, which incorporates rank-aware JS divergence, enabling a multi-faceted measurement of diversity in news recommender systems. However, existing studies mainly focus on short-term recommendation diversity, as the same time we analyze the diversity changes in user-accepted content over the long term, from both the perspectives of intra-list distance [83] and category coverage [31].

Overall, there is a gap in research explicitly addressing the impact of recommendation algorithms on forming information cocoons. We explored the extent to which information cocoons appear in different types of recommender systems and analyzed the algorithmic reasons behind the corresponding results. Addressing the shortcomings of existing simulation methods, we more comprehensively considered real-world factors affecting the diversity of recommended content and proposed a new simulation framework for long-term interactions between users and recommender systems. Since the methods to directly solve the information cocoon problem are currently lacking, we not only explored the effectiveness of representative diversity-oriented recommendation methods in mitigating information cocoons but also proposed a new re-ranking framework to alleviate the homogenization of recommended content.

3 Preliminaries

For ease of understanding, in this section, we summarize the key notations used in the paper in Table 1 and provide a formal definition of the problem.

Item recommendation. In a recommender system, denote the user set as $U = \{u_1, u_2, ..., u_m\}$, the item set as $V = \{v_1, v_2, ..., v_n\}$, and the user click matrix as D. Specifically, D is a binary matrix where $D_{ij} = 1$ indicates user u_i clicked item v_j; $D_{ij} = 0$ indicates no click, and missing values indicate unknown. The recommender RE, based on user features $F_U = \{f_{u_1}, f_{u_2}, ..., f_{u_m}\}$, item features $F_V = \{f_{v_1}, f_{v_2}, ..., f_{v_n}\}$, and user-item interaction data D, predicts the user's preferences for different items. In the top-k recommendation task, the recommender RE recommends k items to each user, providing the recommended item matrix R_U, where $R_U \in R^{m \times k}$.

Table 1 Summary of key notations

Notations	Explanation
U	User set
V	Item set
D	User-item interaction binary matrix
RE	Recommender system
F_U	User feature set
F_V	Item feature set
E_U	User embedding set
E_V	Item embedding set
M_U	Recalled items for all users
$\hat{P}_U$	Predicted click probabilities for recalled items
R_U	Recommended items for all users
L	Users' click item list records
$\hat{p}_{u_i,v_j}$	Predicted click probability of user u_i on item v_j
p_{u_i,v_j}	Simulated click probability of user u_i on item v_j
c_{u_i,v_j}	Binary value of whether user u_i clicks on item v_j
m	The number of users
n	The number of items
t	The number of recommendation iterations
s	The number of items recalled for each user
k	The number of items recommended for each user

Information cocoon detection. In the initial state, the recommender RE makes top-k recommendations for users based on the initial click matrix D. After obtaining the recommended item list R_U, it will be inputted into the click simulation module, where simulates user click behaviors to obtain new user-item interaction data D'. Denote the current recommendation round as i, the historical click data be D_i, and the new interaction data be D'_i. Then, the click matrix D_{i+1} used in the next round is updated via $D_{i+1} = D_i + D'_i$. Iterate the above process t times to obtain the user's click item list $L = \{l_1, l_2, ..., l_t\}$, where l_i represents the user's cumulative click item list after the i-th round of recommendation. After collecting historical interaction data of users at different iterations, it is necessary to conduct further analysis to determine whether the range of items the user is exposed to is gradually narrowing, becoming increasingly limited, thereby posing the risk of falling into an information cocoon.

4 Detection of Information Cocoon

In this section, we first provide an overall introduction to the simulation framework for long-term interactions between users and recommender systems. Following that, we give a formal description of the detailed simulation process. Finally, we explain the key evaluation metrics used for detecting information cocoons.

4.1 Simulation Framework

In order to observe the long-term impact on users under the information filtering effect of recommender systems, we conducted a simulation of multiple rounds of continuous interactions between users and the recommender system. Besides, to investigate the influence of different types of recommendation algorithms on the formation of information cocoons, we utilized a single algorithm-driven recommendation model to interact with users. We repeated experiments with different recommendation algorithms under the same settings. Note that we do not discuss the specific implementation details of the recommendation algorithms, but rather focus on the outer-level interaction between users and the recommendation model.

In each interaction round, the recommendation model was used to generate personalized recommendations for users based on their characteristics and historical behaviors. Then, we employed probability-based click simulations to predict the user's feedback after receiving recommended items. Based on the user's feedback results, we updated the user's historical click data and incrementally retrained the recommendation model. Through iterating this process, we collected data from the long-term interaction between users and the recommender system, analyzing underlying changes in recommended content and user interests.

Figure 2 illustrates the simulation process of one complete recommendation round, which includes the following steps:

1. Feature representation. Train a feature representation model based on user features, item features, and user historical interaction data to obtain embedding representations of users and items.
2. Item retrieval. Perform similarity matching between user embeddings and item embeddings to retrieve a small subset of candidate items that users may be interested in from a large item database.
3. Ranking and recommendation. Utilize a click-through rate (CTR) prediction model to estimate the probabilities of users clicking on the candidate items and select a few items with the highest predicted click probabilities to recommend.
4. User decision. Simulate user behaviors after receiving the recommended items based on the predicted click probabilities, determining whether the user clicks or not and collecting the new user click data.
5. Model and data update. Incrementally update the parameters of the recommendation model using the newly added user interaction data and incorporate the new click data into the historical interaction data, preparing for the simulation of the next recommendation round.

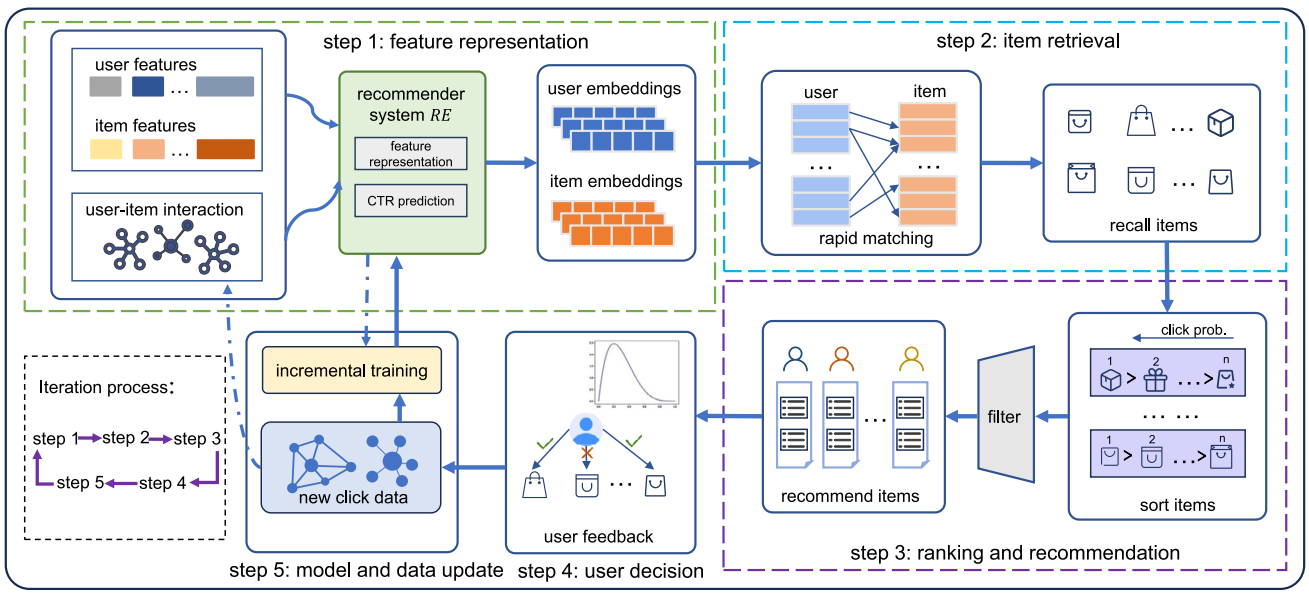


Fig. 2 Recommendation simulation framework

4.2 Formalization of the Simulation Process

As described above, the simulation process can be divided into five stages: feature representation, item retrieval, ranking and recommendation, user decision, as well as model and data update. We next provide detailed explanations for each stage.

Feature representation. Denote the user features as $F_U = \{f_{u_1}, f_{u_2}, ..., f_{u_m}\}$, and the item features as $F_V = \{f_{v_1}, f_{v_2}, ..., f_{v_n}\}$. User features could include attributes such as user ID, user profile, historical click data, etc., while item features could include attributes such as item ID, category, item description, etc. The goal of feature representation learning is to obtain embedding representations of users and items based on their respective features. Firstly, we trained the recommender RE using the user features F_U, item features F_V, and user-item interaction data D as follows:

$$RE \leftarrow \text{Train}_{\text{model}}(F_U, F_V, D). \tag{1}$$

Next, we input the respective features of users and items and the interaction data into the feature representation model RE_{emb} to obtain the embeddings of users and items, as shown below:

$$E_U = RE_{\text{emb}}(F_U, D), \tag{2}$$

$$E_V = RE_{\text{emb}}(F_V, D), \tag{3}$$

where E_U represents the user embeddings with $E_U \in R^{m \times d}$; E_V represents the item embeddings with $E_V \in R^{n \times d}$, where d is the dimension of the embedding vectors.

Item retrieval. In real-world recommender systems, there is a large amount of item data stored, making precise matching time-consuming. Therefore, performing a recall operation to quickly filter out a small number of items that users may be interested in is essential. During the experiment, we utilized Faiss [29], an open-source library developed by Facebook AI Research, for fast recall of candidate items through similarity search. Let's denote the set of recalled items for all users as M_U, where $M_U \in R^{m \times s}$. Here s represents the number of recalled items for each user, typically in the hundreds. For $\forall u_i \in U$, the recalled items M_{u_i} can be obtained through similarity matching between the user embedding E_{u_i} and the item embeddings E_V, as shown below:

$$M_{u_i} = \text{Match}_s(E_{u_i}, E_V). \tag{4}$$

Ranking and recommendation. After obtaining rough candidate items, precise ranking is required to select the items that users are potentially most interested in. Typically, the number of items selected for recommendation falls within the range of tens. During the experiment, we adopted the predicted click probability as the ranking metric and recommended the items with the highest predicted probabilities to users. Specifically, the click-through rate prediction model RE_{CTR} was used to predict the user's click probabilities $\hat{P}_U$ for the recalled items M_U based on user features F_U, item features F_V and user-item interaction data D, as follows:

$$\hat{P}_U = RE_{\text{CTR}}(F_U, F_V, D, M_U), \tag{5}$$

where $\hat{P}_U \in R^{m \times s}$. After that, based on the predicted click probabilities $\hat{P}_U$, we got the top-k recommended items R_U for users, where $R_U \in R^{m \times k}$, as shown below:

$$R_U = \text{Sort}_k(M_U, \hat{P}_U). \tag{6}$$

User decision. After obtaining the item list R_U recommended by the recommender system RE to users, the next step is to simulate the user's behaviors upon receiving these recommended items. Whether a user will click on a recommended item is an uncertain event, which is influenced by multiple factors. For example, when users develop sudden interests in new fields, they may make decisions that deviate from their historical behaviors. Therefore, probabilistic simulation of user decisions is an appropriate approach. Following previous work [7, 37], we sampled a value from a beta distribution with the predicted probability as the mean being the user's true click probability. However, the prediction ability of the recommendation model is limited, and it cannot guarantee that the predicted click probabilities are certainly close to the actual values. Only when the prediction model makes accurate predictions, is the sampled user's true click probability meaningful. Therefore, we multiplied the sampled probability by the model's prediction accuracy to obtain the effective click probability of user u_i on item v_j, as follows:

$$p_{u_i,v_j} = \text{Sample}(\beta(\hat{p}_{u_i,v_j}, \sigma)) \times \text{acc}_{RE}. \tag{7}$$

Then, we sampled a value from the uniform distribution $U(0, 1)$. If this value is less than p_{u_i,v_j}, we consider it as the user did click on the recommended item; otherwise, we consider it as the user refused the recommended item and did not click on it. The process of simulating user u_i clicking on item v_j can be formulated as,

$$c_{u_i,v_j} = \mathbb{1}(p < p_{u_i,vj}), \tag{8}$$

where p is the sampled value from the uniform distribution $U(0, 1)$, and $\mathbb{1}[\cdot] = 1$ if the condition is true, otherwise it is 0.

Model and data update. As users interact continuously with the recommender system, their preferences undergo constant changes. Therefore, it is necessary to update the parameters of the recommendation model in real time. After

each interaction between users and the recommender system, new interaction data is generated. Denote the new interaction data as D', where $D' \in R^{m \times k}$ and $D'_{ij} = c_{u_i, v_j}$. We updated the model parameters of the recommender system using the newly generated training data, as shown below:

$$RE' \leftarrow \text{Train}_{\text{model}}(RE, F_U, F_V, D'). \tag{9}$$

Besides, the interaction data between users and items needs to be updated accordingly. Denote the click list of user u_i in the current h-th recommendation round as $l^h_{u_i}$ and the newly added click item list as Δl_{u_i}, where $\Delta l_{u_i} = \left\{ v_j | D'_{ij} = 1 \right\}$. Then, user u_i's click list $l^{h+1}_{u_i}$ in the next recommendation round is updated by $l^{h+1}_{u_i} = l^h_{u_i} + \Delta l_{u_i}$. Denote the user-item interaction matrix for the $h+1$ recommendation round as D^{h+1}. And for $\forall D^{h+1}_{ij} \in D^{h+1}$, it holds that $D^{h+1}_{ij} = 1$ if item v_j in the click item list $l^{h+1}_{u_i}$, otherwise $D^{h+1}_{ij} = 0$. After completing the update of the model and the interaction data, we started a new round of simulation following the same process. Note that the way the recommendation model adapts to historical and newly added interaction data is determined by the design of the model itself.

By iterating the aforementioned process t rounds, we obtained the clicked item lists of users at different moments, denoted as $L = \left\{ l_1, l_2, ..., l_t \right\}$. By analyzing this data, we can gain deeper insights into the patterns and trends to determine whether users are gradually becoming trapped in an information cocoon. We aim to assess the extent to which users are confined within a limited range of recommended content and the implications of this phenomenon.

4.3 Evaluation Metrics

To assess the degree to which users are confined within homogeneous recommendation content, we introduce two evaluation metrics in this subsection: average relevance distance [83] and normalized category entropy [31]. These metrics are designed to examine the similarity of item content and the diversity of item categories, respectively.

Average relevance distance (ARD). We introduce intra-list distance [83] and measure the homogeneity of recommended items by assessing the similarity of content among those items associated with a particular user. The similarity between items is measured by the cosine value of their embedding vectors. For $\forall u_i \in U$, denoting the click item list as l_{u_i} and the embedding vector of item v_j as E_{v_j}, the user's average relevance distance is calculated by:

$$\text{ARD}_{u_i} = \frac{\sum_{v_i, v_j \in l_{u_i}} (1 - cos(E_{v_i}, E_{v_j}))}{C^2_{|l_{u_i}|}}. \tag{10}$$

The average relevance distance measures the average distance between pairwise items associated with a user, with values ranging from [0, 2], where a smaller value indicates a higher degree of being trapped in an information cocoon.

Normalized category entropy (NCE). Category entropy reflects the evenness of category distribution and can be used to measure the homogeneity of items associated with a user from the perspective of category diversity. Denote the number of items associated with user u_i in category c as n_c, then $p_c = n_c / \sum_c n_c$ and the category entropy equals $-\sum_c p_c \log(p_c)$. Due to the fact that users with more interactive behaviors are generally associated with more diverse content, following the previous work [31, 51], we adopt normalized category entropy to balance this confounding effect. The normalized category entropy of user u_i is defined by:

$$\text{NCE}_{u_i} = \frac{-\sum_c p_c \log(p_c)}{\log(\sum_c \mathbb{1}(p_c > 0))}, \tag{11}$$

where $\mathbb{1}[\cdot]$ has the same meaning as before, i.e., it equals 1 when the condition is true, and 0 otherwise. The value range of normalized category entropy is [0, 1], with a smaller value indicating a more pronounced category preference of user consumption and a higher likelihood of falling into an information cocoon.

After conducting t rounds of interaction simulations between users and the recommender system, we obtained the user click list $L = \left\{ l_0, l_1, l_2, ..., l_t \right\}$, where l_0 is the user's initial click record. Applying the above average relevance distance and normalized category entropy evaluation metrics, we can get the user's distance list $D_{\text{ARD}} = \left\{ d_0, d_1, d_2, ..., d_t \right\}$ and the normalized entropy list $H_{\text{NCE}} = \left\{ h_0, h_1, h_2, ..., h_t \right\}$. If the values of these metrics are gradually decreasing, it means that the information cocoon around the user is gradually forming.

5 Evaluation of Information Cocoon

In this section, we study the issue of the formation of information cocoons. The experiments aim to address the following research questions and reveal our main insights:

- **RQ1**: To what extent do different recommendation algorithms cause information cocoons?
- **RQ2**: What are the potential reasons for recommendation algorithms causing information cocoons?
- **RQ3**: What is the relationship between diversity-oriented recommendation and the formation of information cocoons?

Table 2 Statistics of datasets

Datasets	# users	# items	# train logs	# val. logs	# test logs
MIND	750,434	101,527	2,232,748	188,235	188,236
Music4 All-Onion	53,294	56,488	1,055,859	53,294	53,294

5.1 Experimental Setup

5.1.1 Datasets

We conduct experiments on two publicly available datasets from different fields, and detailed statistics are shown in Table 2. Investigating the information cocoon issue in the fields of news recommendation and music recommendation is important, as users' ideologies, such as cognitive perspectives and emotional states, are significantly influenced by long-term recommended content in these scenarios. The datasets used in the experiments are representative, encompassing rich features such as text, categories, and sentiments.

MIND[1]: MIND [75] is a large-scale news recommendation dataset collected from Microsoft News. In the data we used, there are a total of 2,609,219 behavior log records from 750,434 anonymous users. Each behavior log records the user's click behaviors (click or not) on a group of news displayed on the platform at a certain moment. The news data provided in MIND contains rich information, including topic category, topic subcategory, title, abstract, URL, and entities.

Music4 All-Onion[2]: Music4 All-Onion [47] is a music recommendation dataset extracted from the online music platform Last.fm. For simplicity, we refer to it as Music in the following text. We filtered out 11,500,894 listening records from 53,294 users, ensuring that each user has clicked on at least 30 tracks, and at least 5 clicks are considered valid. Since Music does not contain user behavior logs, we sequentially divided the user-clicked track lists into historical click data and candidate test data based on the click timestamp in ascending order and randomly sampled an equal number of negative samples. Track attributes include lyrics, music genres, and sentiment features of lyrics.

5.1.2 Algorithms

We first explore the impact of three types of recommendation algorithms on the formation of information cocoons. Under each type of recommendation algorithm, we selected some representative recommendation models. Since most state-of-the-art models are improvements and extensions of classical recommendation models, we chose some representative models to investigate the algorithmic causes of information cocoon formation. The experimental models are as follows:

Collaborative filtering-based recommendation algorithms. This type of algorithm makes recommendations based on the click relationship between users and items. **NCF** [23]: Neural collaborative filtering recommendation model. **NGCF** [71]: Graph neural network based collaborative filtering recommendation model.

Content-based recommendation algorithms. This type of algorithm makes recommendations based on users' historical click data and the content of items. **NRMS** [74]: News recommendation model based on neural networks and self-attention mechanism. **NAML** [73]: News recommendation model based on neural networks and multi-view learning. **DKN** [67]: News recommendation model based on neural networks and knowledge graph representation.

Hybrid recommendation algorithms. This type of algorithms combine multiple recommendation algorithms and can incorporate rich features for recommendation. **DeepFM** [21]: Factorization machine and neural networks based recommendation model.

To further explore the relationship between diversity-oriented recommendation methods and the formation of information cocoons, we also introduce some representative diversity-oriented recommendation models, as follows:

DGRec [78]: Diversity-oriented recommendation model based on graph neural networks considering item categories.

DPP [8]: Re-ranking diversity-oriented model based on the determinantal point process.

MMR [6]: Re-ranking diversity-oriented model based on maximal marginal relevance.

5.1.3 Implementation Details

Since different recommendation models have different requirements for input data, we use different data features according to the actual needs of each model for training, which are summarized in Table 3. For music recommendation, we primarily investigate the impact of music lyrics' textual information on the formation of information cocoons. Therefore, we employed content-based news recommendation models for music recommendation. Note that the Music dataset lacks entity information, so we replaced the entities required by the DKN model with sentiment tags extracted from the lyrics. We conducted the experiments on an Intel(R) Xeon(R) Gold 6226R @ 2.90 GHz CPU and two NVIDIA GeForce RTX 3090 GPUs and adopted the deep learning framework Pytorch. For all used base models, we configured the model parameters and training methods,

[1] https://msnews.github.io/.

[2] https://zenodo.org/records/6609677.

Table 3 Training features of different models

Models	MIND	Music
NCF	User ID, news ID, interaction	User ID, track ID, interaction
NGCF	User ID, news ID, interaction	User ID, track ID, interaction
NRMS	User ID, news ID, title, interaction	User ID, track ID, lyrics, interaction
NAML	User ID, news ID, title, abstract, topic category and subcategory, interaction	User ID, track ID, lyrics, genre, interaction
DKN	User ID, news ID, title, entities in title, interaction	User ID, track ID, lyrics, sentiment, interaction
DeepFM	User ID, news ID, topic category and subcategory, interaction	User ID, track ID, genre, interaction
DGRec	User ID, news ID, topic category, interaction	User ID, track ID, genre, interaction

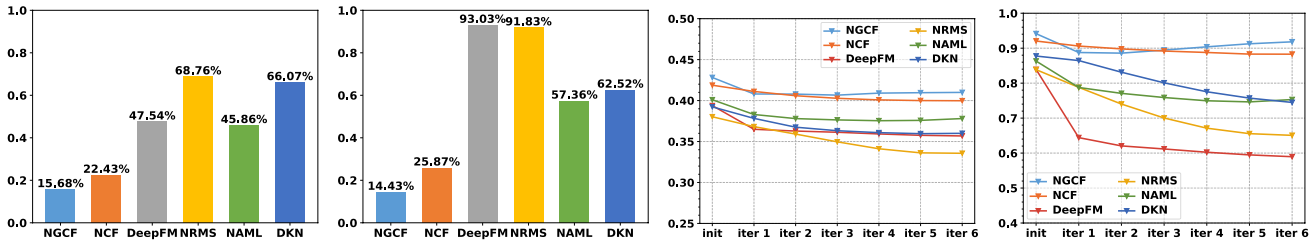
including the dimension of embedding vectors, according to the corresponding official suggestions. We simulated the behaviors of users from the test set and conducted a total of 6 rounds of interaction simulations. In each simulation round, the number of recalled items is 500, and the number of recommended items for users is 50. We estimate the accuracy of the recommendation model based on its performance on the test data. In addition, there is an uneven distribution of news categories in MIND. Therefore, we sampled 51,029 news as the candidate news set. Furthermore, to make the experiments reproducible, we make the code publicly available at https://anonymous.4open.science/r/information_cocoon_study-945E.

5.2 Experimental Results

5.2.1 RQ1: To What Extent do Different Recommendation Algorithms Cause Information Cocoons?

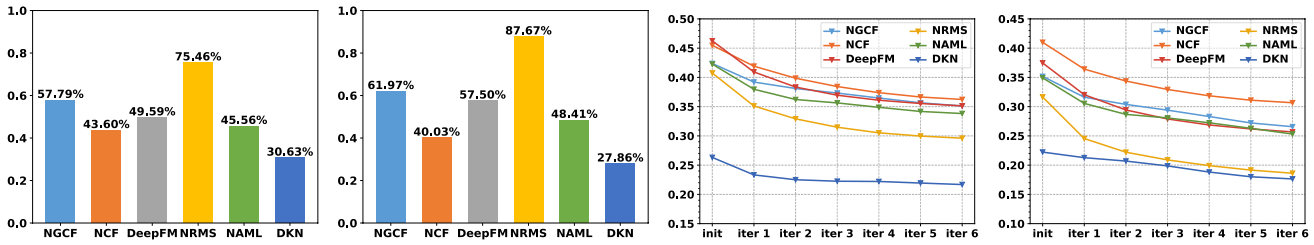
We analyzed 6 consecutive rounds of interaction data of users who initially clicked on at least 10 items. We focus on the textual content (ARD_{text}) changes of user-clicked items, i.e., news text and track lyrics. Besides, considering the respective characteristics of the two datasets, we also pay attention to the changes in topic categories of news (NCE) and sentiment content of tracks (ARD_{sent}). According to the definitions in Sect. 4.3, the homogeneity of items is negatively correlated with ARD and NCE. The decline in the metric values indicates that the content users consume tends to be homogeneous. And the more pronounced the downward trend, the higher the risk of users falling into an information cocoon. Under the long-term influence of different recommendation models, the statistics of the percentage of users experiencing a decline in these metrics are shown in Figs. 3a, b and 4a, b. To better observe the changes in these metrics throughout the recommendation process, we calculated the average metric values for these users encountering homogenized recommendations, as illustrated in Figs. 3c, d and 4c, d, with the detailed data provided in Tables 4 and 5.

According to the results, different recommendation models expose users to varying degrees of the risk of being trapped in information cocoons. On the MIND dataset, collaborative filtering-based recommendation models NGCF and NCF cause a relatively small decline and downward trend in the ARD_{text} and NCE metrics; However, content-based recommendation model NRMS and hybrid recommendation model DeepFM show the most significant decline impact on the ARD_{text} and NCE metrics, respectively; Content-based recommendation models NAML and DKN have similar moderate performance. On the Music dataset, NGCF and NCF show a noticeable increase in the portion of decline in the ARD_{text} and ARD_{sent} metrics, but the downward trend remains relatively mild; NRMS still have a significant impact, causing the largest portion



(a) change rate (ARD_{text}) (b) change rate (NCE) (c) avg. change (ARD_{text}) (d) avg. change (NCE)

Fig. 3 The changes in ARD_{text} and NCE metrics of user-clicked news on MIND. ARD_{text} is the averaged relevance distance of the news text. NCE is the normalized category entropy of the news topics



(a) change rate (ARD_{text}) (b) change rate (ARD_{sent}) (c) avg. change (ARD_{text}) (d) avg. change (ARD_{sent})

Fig. 4 The changes in ARD_{text} and ARD_{sent} metrics of user-clicked tracks on Music. ARD_{text} and ARD_{sent} are the averaged relevance distances of lyric text and sentiment, respectively



(a) initial status (b) iteration 2 (c) iteration 4 (d) iteration 6

Fig. 5 Evolution of word clouds for a random user's historical clicked news under NRMS's influence

Table 4 The average results for users receiving homogenized content on MIND

Models	ARD_{text}							NCE						
	Init	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5	Iter 6	Init	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5	Iter 6
NGCF	0.4282	0.4080	0.4079	0.4066	0.4091	0.4096	0.4100	0.9415	0.8877	0.8858	0.8946	0.9039	0.9126	0.9182
NCF	0.4188	0.4112	0.4059	0.4028	0.4008	0.4000	0.3998	0.9205	0.9061	0.8980	0.8918	0.8877	0.8831	0.8828
DeepFM	0.3939	0.3649	0.3628	0.3613	0.3592	0.3576	0.3568	0.8392	0.6442	0.6206	0.6118	0.6024	0.5948	0.5897
NRMS	0.3802	0.3681	0.3591	0.3496	0.3411	0.3362	0.3356	0.8386	0.7880	0.7402	0.7003	0.6712	0.6556	0.6507
NAML	0.4010	0.3830	0.3780	0.3764	0.3754	0.3758	0.3780	0.8631	0.7876	0.7708	0.7589	0.7496	0.7464	0.7531
DKN	0.3923	0.3783	0.3675	0.3632	0.3608	0.3596	0.3600	0.8775	0.8646	0.8313	0.8009	0.7754	0.7572	0.7449

Table 5 The average results for users receiving homogenized content on music

Models	ARD_{text}							ARD_{sent}						
	Init	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5	Iter 6	Init	Iter 1	Iter 2	Iter 3	Iter 4	Iter 5	Iter 6
NGCF	0.4238	0.3920	0.3810	0.3732	0.3647	0.3565	0.3512	0.3517	0.3162	0.3038	0.2940	0.2831	0.2721	0.2657
NCF	0.4548	0.4191	0.3986	0.3844	0.3739	0.3665	0.3623	0.4100	0.3641	0.3438	0.3294	0.3185	0.3110	0.3067
DeepFM	0.4625	0.4093	0.3835	0.3696	0.3610	0.3554	0.3513	0.3747	0.3204	0.2943	0.2789	0.2687	0.2619	0.2568
NRMS	0.4074	0.3512	0.3291	0.3148	0.3052	0.2997	0.2961	0.3165	0.2455	0.2221	0.2089	0.1992	0.1915	0.1862
NAML	0.4227	0.3797	0.3622	0.3564	0.3488	0.3418	0.3382	0.3492	0.3055	0.2868	0.2807	0.2724	0.2629	0.2531
DKN	0.2632	0.2333	0.2251	0.2225	0.2221	0.2194	0.2169	0.2223	0.2128	0.2070	0.1987	0.1882	0.1801	0.1764

of decline and relatively pronounced downward trends in both metrics, while DeepFM shows a clear weakening of influence; Although DKN causes lower decline ratios on the two metrics compared to other models, it significantly increases the homogeneity of lyrics text and sentiment content received by users.

We further analyze the underlying reasons for the different performances of some recommendation models on the two datasets. NGCF and NCF primarily rely on

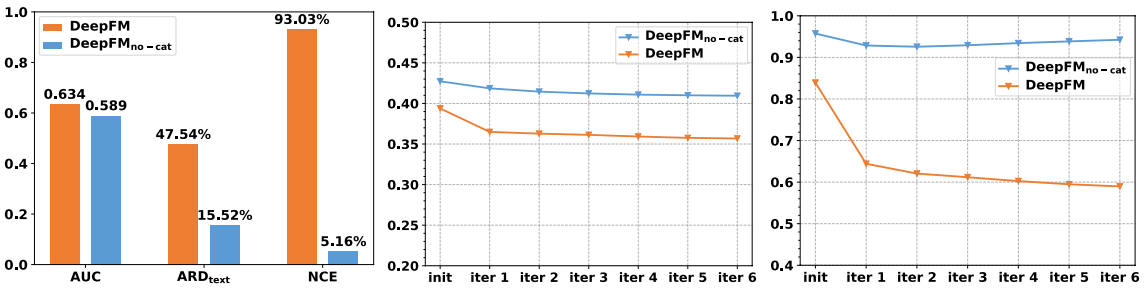
collaborative information from user-item interactions for recommendations. The interaction sparsity rates of MIND and Music datasets are 0.024% and 0.345%, respectively. Therefore, on the denser Music dataset, NGCF and NCF better learn users' music content preferences, recommending tracks that are more similar to their historical consumption. The influence of DeepFM on the Music dataset is significantly less pronounced compared to its impact on the MIND dataset. The main reason is that music genres, as one of the important features learned by DeepFM, are associated with various factors such as the rhythm and melody of songs. In this case, DeepFM failed to capture additional insights into users' preferences for music content. Additionally, for music recommendation, we replaced the entity features required by DKN with sentiment features, resulting in differences from news recommendation results.

Thus, we can glean crucial insights from the above results. Collaborative filtering-based recommendation algorithms cause a lighter degree of information cocoons. The main reason is that, in addition to recommending content similar to users' historical consumption, it also discovers similarities between users based on their behaviors, thereby exposing them to some new content. Besides, NRMS, DeepFM on MIND, and DKN on Music exhibit the most significant performance in narrowing down the scope of textual content, topic categories, and sentiment content of recommended items, respectively. This is because these models, in addition to learning the user-item interaction features, only leverage relatively simple additional training features. After capturing users' content preferences in specific aspects, they tend to provide users with excessively similar items. Furthermore, NAML, which utilizes rich training features including user-item interaction, item text, and category, does not lead to the most severe information cocoons. We infer that when the model captures user preferences from multiple aspects, its ability to recommend a more diverse range of items could be enhanced.

Visualization. To intuitively observe the changes in the items that users receive during the interaction with the recommender system, using news recommendations as an example, we visualized the word clouds of the historical click news of a randomly selected user in different periods. Figure 5 shows the word cloud evolution of the clicked news by a user under the influence of NRMS. From Fig. 5a, we know that the user was interested in multiple topics at the start, such as "lifestyle," "entertainment," and "crime." However, as shown in Fig. 5b, the topics of the user's clicked news became narrow after the 2-nd round of recommendation, with a majority being crime-related news. Figures 5c and d show that the keywords "police" and "newscrime" became more and more obvious, and new keywords under the same topic, such as "murder" and "crash," were constantly appearing. Therefore, we can conclude that the user gradually fell into an information cocoon, being affected by the recommendation algorithm.

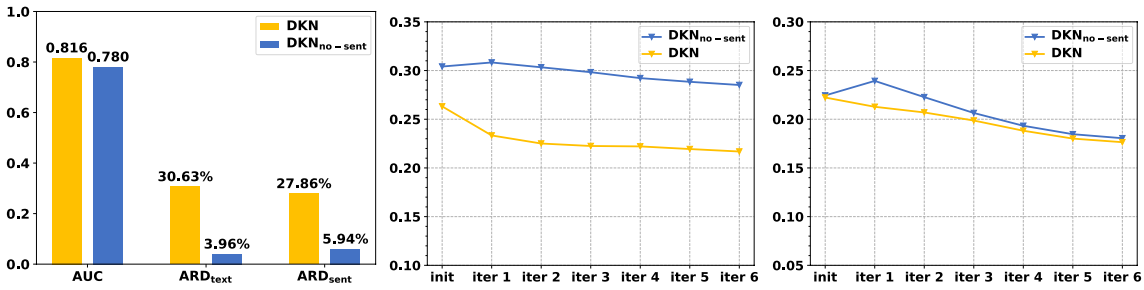
5.2.2 RQ2: What are the Potential Reasons for Recommendation Algorithms Causing Information Cocoons?

In the above experiments, besides focusing on the changes in the textual content of recommended items, we also examined the changes in news categories and lyrics sentiment based on the respective characteristics of the two datasets. The results show that DeepFM on the MIND dataset and DKN on the Music dataset behave significant information cocoon effects. Based on this observation, we conducted further exploration of these two models (hereinafter, we simplify them as DeepFM and DKN for brevity). Specifically, for DeepFM, we removed the training features related to news categories and denoted the obtained new model as $\text{DeepFM}_{\text{no-cat}}$. F or DKN, we excluded the training features related to tracking sentiment and denoted the resulting model as $\text{DKN}_{\text{no-sent}}$. Under such a setup, the models reduce reliance on the specific content of items and focus more on the user-item interaction information. Following the same settings as the previous experiments, we simulated multiple rounds



(a) accuracy and change rate (b) avg. change (ARD_{text}) (c) avg. change (NCE)

Fig. 6 The effect of the DeepFM models on the information cocoon on MIND



(a) accuracy and change rate (b) avg. change (ARD_{text}) (c) avg. change (ARD_{sent})

Fig. 7 The effect of the DKN models on the information cocoon on Music

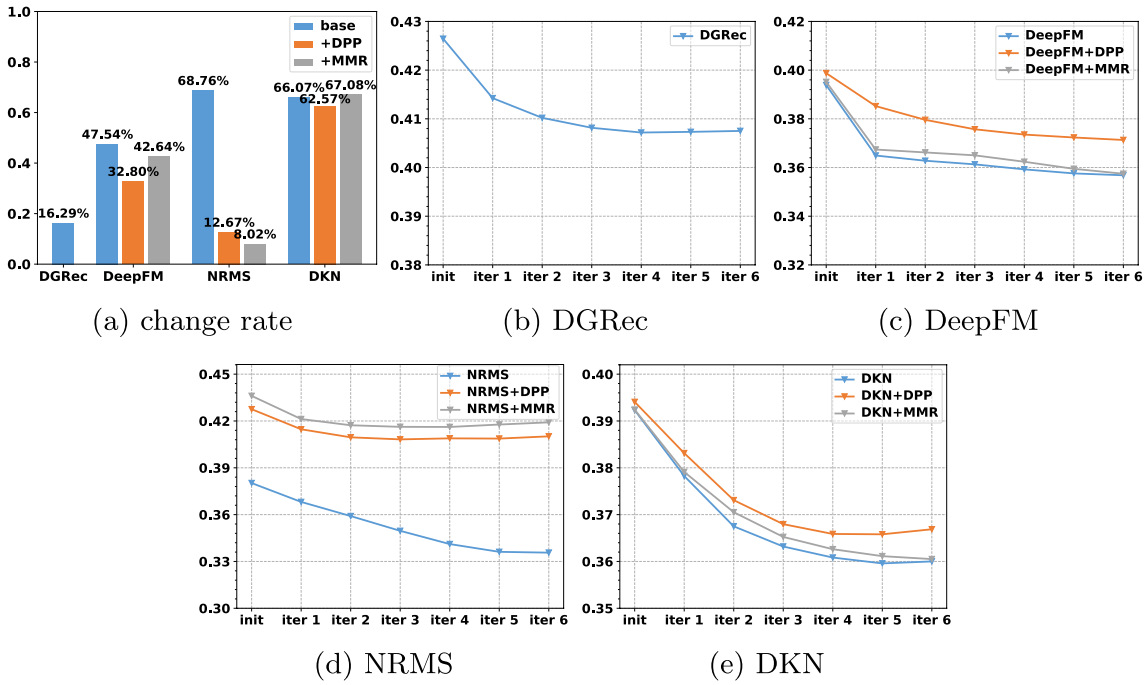
of interactions between users and the two recommendation model variants, with the results shown in Figs. 6 and 7.

As shown in Fig. 6, DeepFM$_{no\text{-}cat}$ causes 15.52% and 5.16% of users experiencing decline in ARD$_{text}$ and NCE metrics, respectively, which is much less than the case on DeepFM. It can also be observed that under the influence of DeepFM$_{no\text{-}cat}$, the decline in values for ARD$_{text}$ and NCE metrics are milder, with a much more gradual downward trend. Similarly, DKN$_{no\text{-}sent}$ only leads to a decrease in less than 6% of users on both ARD$_{text}$ and ARD$_{sent}$ metrics. The extent of the decline in these metric values is less pronounced than in the DKN situation. In comparison to the original models, DeepFM$_{no\text{-}cat}$ and DKN$_{no\text{-}sent}$ only lack category information and sentiment content of items in training features, respectively. However, this results in significant differences in the formation of information cocoons. This validates our previous inference that when the model excessively learns users' preferences in certain specific aspects, it tends to recommend an excess of items with similar characteristics, thereby propelling the formation of information cocoons. Furthermore, we also observe that introducing additional effective training features can enhance the accuracy of recommendations but unintentionally deepens users' immersion in information cocoons. This implies a risk that users may gradually become immersed in their surrounding information cocoons.

5.2.3 RQ3: What is the Relationship Between Diversity-Oriented Recommendation and the Formation of Information Cocoons?

Diversity-oriented recommendation is beneficial for enhancing the diversity of recommended item lists, theoretically mitigating the issue of an information cocoon. To further explore whether diversity-oriented recommendation models may cause information cocoons as well, and the extent of the impact, we also simulated the long-term interactions between users and diversity-oriented recommendation models. We conducted the experiments based on one in-processing graph model: DGRec [78] and two post-processing re-ranking models: DPP [8] and MMR [6]. The post-processing models usually enhance the diversity of recommendations in the ranking stage, thus allowing for flexible integration with existing recommendation models. In our experiments, we investigated the combined effects of the two post-processing models with DeepFM, NRMS, and DKN models, which are known to exacerbate information cocoons. During this process, we utilized the item embeddings learned by the models to obtain item similarities and used the predicted click probabilities as item scores. We report the experimental results on the MIND dataset (similar results were observed on the Music dataset), as shown in Fig. 8.

From the results, it can be observed that the end-to-end diversity-oriented graph model DGrec leads to 16.29% of users receiving narrower news content, and the declining trend of ARD$_{text}$ is more gradual compared to most other models. For post-processing models DPP and MMR, when combined with the NRMS model, both show a significant alleviation of information cocoons; when combined with the DeepFM model, DPP exhibits a certain degree of mitigating effect, while MMR's alleviation is somewhat weaker; when combined with the DKN model, DPP shows a weak mitigating effect, while MMR has little to no effect. Therefore, we conclude that although diversity-oriented recommendation models, in most cases, can alleviate the formation of information cocoons by increasing the diversity of the recommended lists to some extent, they have lots of room to improve. Additionally, we infer that the differing performance of DPP and MMR when combined with different recommendation models is attributed to variations in representation learning and prediction among the different recommendation models. Overall, more efforts need to be made to alleviate the information cocoons.



(a) change rate (b) DGRec (c) DeepFM

(d) NRMS (e) DKN

Fig. 8 The effect of the diverse recommendation models on the decrease of the ARD_{text} metric on MIND

6 Mitigate the Information Cocoon

Through the above experiments, we found that under the influence of different recommendation algorithms, users fall into different degrees of information cocoons. Considering the limitations of diversity-oriented recommendation models in alleviating information cocoons, we propose an Information Cocoon Mitigation Framework (ICMF). Applied during the re-ranking stage, ICMF can also be flexibly integrated with existing recommendation models. Noting that diversity-oriented recommendation models are typically helpful in enhancing the diversity of the recommendation list, but there may be cases where the items in the recommendation list are diverse from each other, yet similar to the user's historically consumed items. Therefore, from the perspective of improving the diversity of user-consumed content, we propose ICMF. The formal descriptions are as follows:

For user u_i, let the historical click list be denoted as l_{u_i}. Based on the embedding vectors of items, we perform an average operation to obtain the representation of the user's historical consumption content, as follows:

$$C_{u_i} = \frac{1}{|l_{u_i}|} \sum_{v_j \in l_{u_i}} E_{v_j}. \tag{12}$$

Next, the similarity between candidate item v_j and the user's historical consumption content is defined as:

$$\text{sim}(u_i, v_j) = \cos(C_{u_i}, E_{v_j}). \tag{13}$$

Then, we denote the predicted probability of user u_i clicking on item v_j as $\text{prob}(u_i, v_j)$, and the comprehensive score of candidate item v_j is calculated by:

$$S_{ij} = \text{prob}(u_i, v_j) - (1 - \lambda) \cdot \text{sim}(u_i, v_j), \tag{14}$$

where $\lambda \in [0, 1]$, and is used to balance recommendation accuracy and diversity, with a smaller value corresponding to a greater emphasis on diversity.

After the recommender system completes the recall of candidate items, we calculate the comprehensive scores of candidate items based on the predicted click probabilities and the similarities between candidate items and the user's historical consumption content. We recommend items with the highest comprehensive scores to users to balance the accuracy and diversity of recommendations. According to the model definition, we know that ICMF is independent of the base recommendation model and contains only one parameter, λ. In the following short-term recommendation experiments, we demonstrate how λ affects recommendation accuracy and diversity. It is suggested to determine the appropriate value of λ based on the short-term recommendation results, ensuring that a considerable recommendation accuracy is maintained while achieving a significant advantage in recommendation diversity.

Efficiency analysis. Let the length of user u_i's historical click list be p, the number of candidate items be s, and the

number of recommended items be k. In the case of using heap sort, the time complexity of the entire re-ranking stage is $O(p + 2s + s \cdot log(k)) \approx O(s \cdot log(k))$, which is time efficient.

7 Evaluation of ICMF

In this section, we study the mitigation of information cocoons. Through conducting experiments, we address the following research question:

- **RQ4**: How effective is the proposed information cocoon mitigation framework ICMF in mitigating information cocoons?

7.1 Experimental Setup

To evaluate the accuracy (by recall@5) of ICMF, we conducted experiments on the user behavior data from the test set. We used items with "click" or "not click" labels in the user behavior logs as candidate items and employed ICMF for item re-ranking to provide recommendations to users. For each used behavior log, we ensure that the number of clicked items by the user is no less than 10, and the number of positive candidate items is no less than 5. Besides, under the same settings as the above experiments, we simulated long-term interactions between these users and the recommender system integrated with ICMF to assess its effectiveness in mitigating information cocoons.

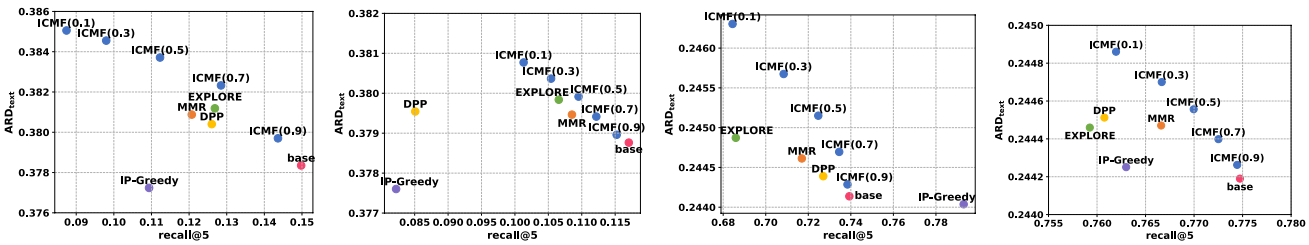
We conducted experiments using NRMS and DKN as the base models. The competitive re-ranking methods include two classical diversity-oriented methods **DPP** [8] and **MMR** [6] (see Sect. 5.1.2) and two latest methods: **IP-Greedy** [25]: A diversity-aware method for maximum inner product search; **EXPLORE** [12]: A recommendation strategy for maximizing diversity exploration. For all methods, we refer to the official instructions carefully configured and tuned to achieve the desired performance.

7.2 Experimental Results

7.2.1 RQ4: How Effective is the Proposed Information Cocoon Mitigation Framework ICMF in Mitigating Information Cocoons?

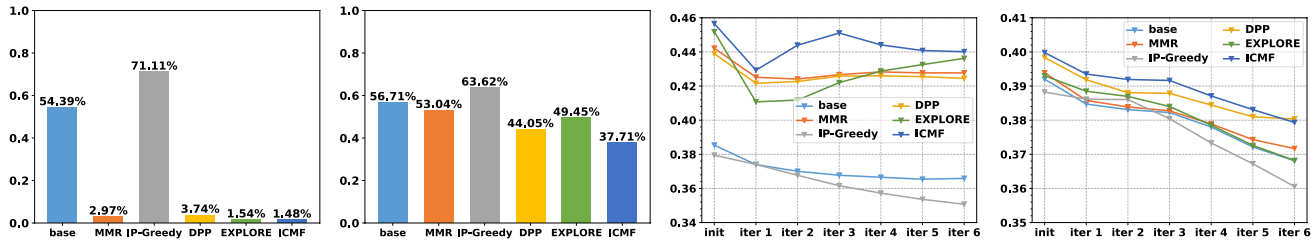
Single-round recommendation results. We provided the top 5 recommendations for users, calculating both the average relevance distance between the recommended items and the user's historically clicked items and the recommendation recall. Figure 9 illustrates the scatter plots for recall@5 and ARD_{text} with $\lambda = [0.1, 0.3, 0.5, 0.7, 0.9]$. It can be observed that as the value of λ decreases, with a greater emphasis on recommendation diversity, the recommended items received by users diverge more from their historical consumption content, but accuracy also decreases accordingly. Besides, the base model achieves almost the best accuracy and the worst diversity. These show a trade-off relationship between recommendation accuracy and diversity. Overall, the points of ICMF are located at the upper right of the points of other diversity-aware methods, demonstrating that ICMF can better balance the accuracy and diversity of recommendations.

Long-term recommendation results. Based on the results of the single-round recommendation, for the NRMS and DKN models, we set λ to 0.5 and 0.1, respectively. This allows ICMF to maintain comparable accuracy to other methods in most cases while having a noticeable advantage in the ARD_{text} value. We conducted the simulations of long-term recommendation and tracked the changes in the text features of the clicked items by users, as shown in Figs. 10 and 11. The results show that compared to other methods, ICMF can effectively reduce the proportion of users experiencing a decline in the ARD_{text} metric. For the average changes in the ARD_{text} metric, in most cases, the line representing ICMF is above those of the other methods, demonstrating a better alleviation of information cocoons. Besides, the difference in results before and after applying the ICMF module to the base model also proves its effectiveness. Compared to the base model, some re-ranking methods such as dpp can help alleviate the information



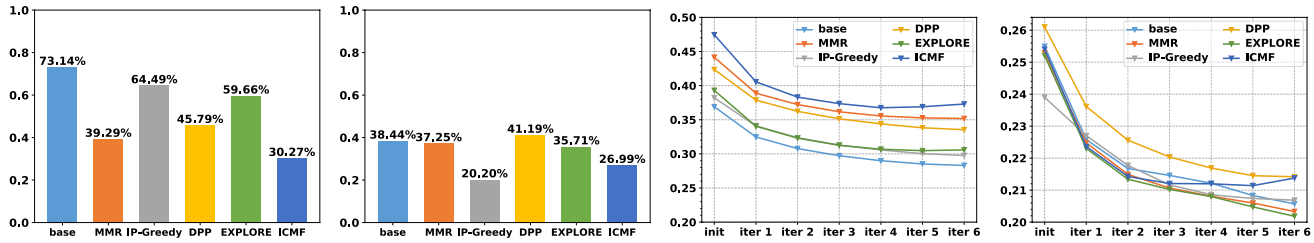
(a) NRMS on MIND (b) DKN on MIND (c) NRMS on Music (d) DKN on Music

Fig. 9 Single-round recommendation results of ICMF with $\lambda = [0.1, 0.3, 0.5, 0.7, 0.9]$



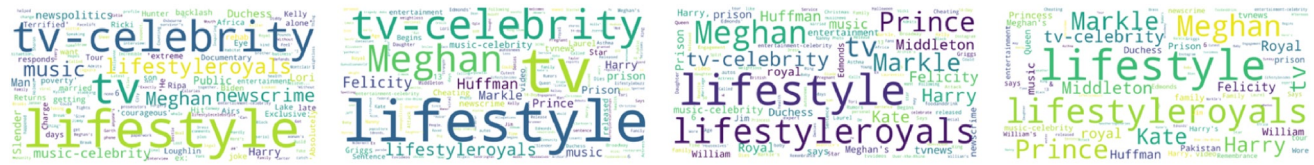
(a) change rate (NRMS) (b) change rate (DKN) (c) avg. change (NRMS) (d) avg. change (DKN)

Fig. 10 The long-term recommendation effect on the decrease of the ARD_{text} metric on MIND



(a) change rate (NRMS) (b) change rate (DKN) (c) avg. change (NRMS) (d) avg. change (DKN)

Fig. 11 The long-term recommendation effect on the decrease of the ARD_{text} metric on Music



(a) initial status (b) iteration 2 (c) iteration 4 (d) iteration 6

Fig. 12 Evolution of word clouds for a random user's historical clicked news under NRMS's influence



(a) initial status (b) iteration 2 (c) iteration 4 (d) iteration 6

Fig. 13 Evolution of word clouds for a random user's historical clicked news under the influence of NRMS and ICMF

cocoon, while some may even exacerbate it. Moreover, under the influence of ICMF, the ARD_{text} values exhibit a trend of initially decreasing and then increasing on the Music dataset, indicating that ICMF has more potential to alleviate the information cocoon in long-term recommendations.

Visualization. To intuitively observe the ability of ICMF to mitigate the homogenization of recommended content, we compared the changes in the content received by users during interactions with the recommender system before and after using ICMF. Similar to before, using a news recommendation task based on the NRMS model as an example, we visualized word clouds of news clicked by a randomly selected user at different periods. The results are shown in Figs. 12 and 13. In the initial state, the user's main interests were concentrated on topics such as "TV," "TV-celebrity," and "lifestyle." As shown in Fig. 12, the NRMS-based recommendation model, after capturing the user's initial interests, continued to recommend

similar content, leading to the repeated reinforcement of topics like "TV-celebrity" and "lifestyle." As depicted in Fig. 13, under the combined influence of NRMS and ICMF, the recommendation model not only recommended content that matched the user's initial interests but also gradually introduced new topics such as "weather" and "sports," preventing the user from repeatedly encountering homogeneous content. The visualization results suggest that ICMF helps mitigate the information cocoon effect.

8 Discussion

In this section, we summarize the experimental findings and state the limitations of the current work and future directions.

In this study, we explore the impact of recommendation algorithms on the formation of information cocoons by conducting interactive simulation experiments between users and recommendation systems. The results indicate that different recommendation algorithms can trap users into varying degrees of information cocoons. We analyzed the reason behind this, which is that when the recommendation models have learned the consumption preferences of users in certain aspects, the models tend to recommend more similar items to cater to user preferences, thereby trapping users in the information cocoon. Besides, the experimental results also provide some insights for mitigating the information cocoon: the recommendation models should mine a wider range of potential user interests, such as recommending based on similar users or social networks; the recommendation models should learn from multiple sources of information and capture user preferences as comprehensively as possible, rather than relying solely on specific training features. In addition, the experimental results conducted on diversity-oriented models show that, in most cases, diversity-oriented recommendations can partially alleviate information cocoons. However, the effectiveness of diversity-aware re-ranking methods is limited by the learning capability of the base recommendation model. Existing work has not yet fully resolved the issue of information cocoons.

This work still has some limitations. The real-world recommendation environment is complex. Different recommendation algorithms are customized based on various data characteristics, recommendation scenarios, and user contexts, with recommendation platforms also imposing policy constraints and ethical norms. Integrating all these factors into a unified framework is challenging. Whereas this study begins a preliminary exploration of the information cocoon phenomenon from a generalized perspective, there is room for improvement in the simulation method. Since the simulation method impacts on the experimental results, it should be improved and personalized to match specific recommendation scenarios. Online experiments can also be conducted simultaneously to validate the reliability of the simulation method. Besides, although the proposed framework ICMF helps to mitigate information cocoons, there is room for improvement. In the future, it is encouraged to develop new recommendation algorithms to help eliminate or minimize the possibility of information cocoons.

Overall, future work includes these areas: (1) Improve the simulation framework according to actual needs to explore the phenomenon of information cocoons in more specific recommendation scenarios. (2) Validating the current simulation with online tests would be meaningful, but it requires the operational real-world recommendation environment and substantial human resources, which presents challenges for researchers. (3) Exploring effective methods to detect, alleviate, and break the information cocoon, preventing users from being trapped in the information cocoon. (4) Proactively identify users who are already trapped or being trapped in the information cocoon and take active measures to help them break free from it.

9 Conclusion

In this work, we collected user behavior data by conducting extensive simulation experiments, exploring whether an information cocoon forms during the recommendation process and analyzing the key factors leading to the information cocoon from the perspective of algorithms. To address the current research gap in mitigating information cocoons, we also proposed a lightweight diversity-aware re-ranking framework ICMF and demonstrated its effectiveness through comparative experiments with state-of-the-art methods.

Acknowledgements This work was supported in part by NSFC (National Science Foundation of China) [Grant Numbers 62172237, U1936206, and 62077031]. This work was also partially supported by Tianjin postgraduate research and innovation project [Grant Numbers 2022BKYZ040].

Authors' contribution LZ: data curation, formal analysis, methodology, software, investigation, visualization, writing—original draft, writing—review and editing. YL: data curation, software, visualization, writing—original draft, funding acquisition. HW: resources, writing—original draft. CS: conceptualization, methodology, supervision, project administration, writing—review and editing, funding acquisition. XY: supervision, project administration, funding acquisition.

Funding National Science Foundation of China [Grant Numbers 62172237, U1936206, and 62077031]. Tianjin postgraduate research and innovation project [Grant Numbers 2022BKYZ040].

Availability of data and materials The datasets used in our experiments are publicly accessible, which have been described in detail in Section 5.1.1 in our manuscript.

Declaration

Competing interests As the authors of this manuscript "An Exploratory Study on Information Cocoon in Recommender Systems", we hereby declare that we have no financial or non-financial interests that could be perceived as influencing the work reported in this paper. We affirm that there are no competing interests relevant to the content of this paper. Furthermore, we have no affiliations with or involvement in any organization or entity that has a conflicting interest in the findings, materials, or conclusions presented in this manuscript. We also confirm that we do not hold any financial or proprietary interests related to the subject matter discussed in this paper.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Anderson A, Maystre L, Anderson I et al (2020) Algorithmic effects on the diversity of consumption on Spotify. In: Proceedings of the web conference, pp 2155–2165
2. Areeb QM, Nadeem M, Sohail SS et al (2023) Filter bubbles in recommender systems: fact or fallacy—a systematic review. WIREs Data Mining Knowl Discov 13(6):e1512
3. Aridor G, Gonçalves D, Sikdar S (2020) Deconstructing the filter bubble: user decision-making and recommender systems. In: Fourteenth ACM conference on recommender systems, pp 82–91
4. Bobadilla J, Ortega F, Hernando A et al (2013) Recommender systems survey. Knowl Based Syst 46:109–132
5. Bountouridis D, Harambam J, Makhortykh M et al (2019) SIREN: a simulation framework for understanding the effects of recommender systems in online news environments. In: Proceedings of the conference on fairness, accountability, and transparency. ACM, pp 150–159
6. Carbonell JG, Goldstein J (2017) The use of MMR, diversity-based reranking for reordering documents and producing summaries. SIGIR Forum 51(2):209–210
7. Chaney AJB, Stewart BM, Engelhardt BE (2018) How algorithmic confounding in recommendation systems increases homogeneity and decreases utility. In: Proceedings of the 12th ACM conference on recommender systems, pp 224–232
8. Chen L, Zhang G, Zhou E (2018) Fast greedy MAP inference for determinantal point process to improve recommendation diversity. In: Advances in neural information processing systems, pp 5627–5638
9. Cheng P, Wang S, Ma J et al (2017) Learning to recommend accurate and diverse items. In: Proceedings of the 26th international conference on world wide web. ACM, pp 183–192
10. Chitra U, Musco C (2020) Analyzing the impact of filter bubbles on social network polarization. In: The thirteenth ACM international conference on web search and data mining, pp 115–123
11. Cinelli M, Morales G, Galeazzi A et al (2021) The echo chamber effect on social media. Proc Natl Acad Sci USA 118(9):e2023301118
12. Coppolillo E, Manco G, Gionis A (2024) Relevance meets diversity: a user-centric framework for knowledge exploration through recommendations. In: Proceedings of the 30th ACM SIGKDD conference on knowledge discovery and data mining. ACM, pp 490–501. https://doi.org/10.1145/3637528.3671949
13. Donkers T, Ziegler J (2021) The dual echo chamber: modeling social media polarization for interventional recommending. In: Fifteenth ACM conference on recommender systems, pp 12–22
14. Du S, Gregory S (2016) The echo chamber effect in twitter: does community polarization increase? In: Complex networks & their applications V—proceedings of the 5th international workshop on complex networks and their applications, vol 693. Springer, pp 373–378
15. Flaxman S, Goel S, Rao JM (2016) Filter bubbles, echo chambers, and online news consumption. Public Opin Q 80(S1):298–320
16. Gan L, Nurbakova D, Laporte L, et al (2020) Enhancing recommendation diversity using determinantal point processes on knowledge graphs. In: Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval, pp 2001–2004
17. Gao Z, Shen T, Mai Z et al (2022) Mitigating the filter bubble while maintaining relevance: targeted diversification with vae-based recommender systems. In: The 45th international ACM SIGIR conference on research and development in information retrieval, pp 2524–2531
18. Ge Y, Zhao S, Zhou H et al (2020) Understanding echo chambers in e-commerce recommender systems. In: Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval, pp 2261–2270
19. Geschke D, Lorenz J, Holtz P (2019) The triple-filter bubble: using agent-based modelling to test a meta-theoretical framework for the emergence of filter bubbles and echo chambers. Br J Soc Psychol 58(1):129–149
20. Gillani N, Yuan A, Saveski M et al (2018) Me, my echo chamber, and I: introspection on social media polarization. In: Proceedings of the world wide web conference. ACM, pp 823–831
21. Guo H, Tang R, Ye Y et al (2017) Deepfm: a factorization-machine based neural network for CTR prediction. In: Proceedings of the twenty-sixth international joint conference on artificial intelligence, pp 1725–1731
22. Han H, Wang C, Zhao Y et al (2022) SSLE: a framework for evaluating the "filter bubble'' effect on the news aggregator and recommenders. World Wide Web 25(3):1169–1195
23. He X, Liao L, Zhang H et al (2017) Neural collaborative filtering. In: Proceedings of the 26th international conference on world wide web, pp 173–182
24. He Y, Zou H, Yu H et al (2019) Diversity-aware recommendation by user interest domain coverage maximization. In: IEEE ICDM, pp 1084–1089
25. Hirata K, Amagata D, Fujita S et al (2022) Solving diversity-aware maximum inner product search efficiently and effectively. In: Sixteenth ACM conference on recommender systems, pp 198–207
26. Huang Y, Wang W, Zhang L et al (2021) Sliding spectrum decomposition for diversified recommendation. In: The 27th ACM SIGKDD conference on knowledge discovery and data mining, pp 3041–3049
27. Isinkaye FO, Folajimi YO, Ojokoh BA (2015) Recommendation systems: principles, methods and evaluation. Egypt Inform J 16(3):261–273
28. Jiang R, Chiappa S, Lattimore T et al (2019) Degenerate feedback loops in recommender systems. In: Proceedings of the AAAI/ACM conference on AI, ethics, and society, pp 383–390

29. Johnson J, Douze M, Jégou H (2021) Billion-scale similarity search with GPUs. IEEE Trans Big Data 7(3):535–547
30. Kim J, Jeon H, Lee J et al (2023) Diversely regularized matrix factorization for accurate and aggregately diversified recommendation. In: Advances in knowledge discovery and data mining—27th Pacific-Asia conference on knowledge discovery and data mining, lecture notes in computer science, vol 13937. Springer, pp 361–373
31. Li N, Gao C, Piao J et al (2022) An exploratory study of information cocoon on short-form video platform. In: Proceedings of the 31st ACM international conference on information & knowledge management, pp 4178–4182
32. Li N, Ban X, Ling C et al (2024) Modeling user fatigue for sequential recommendation. In: SIGIR, pp 996–1005
33. Li Z, Xie Y, Zhang WE et al (2024) Disentangle interest trend and diversity for sequential recommendation. Inf Process Manag 61(3):103619
34. Liang S, Alimu N, Si H et al (2023) Influence of artificial intelligence recommendation on consumers' purchase intention under the information cocoon effect. In: HCI in business, government and organizations—10th international conference, lecture notes in computer science, vol 14038. Springer, pp 249–259
35. Liang Y, Qian T, Li Q et al (2021) Enhancing domain-level and user-level adaptivity in diversified recommendation. In: The 44th international ACM SIGIR conference on research and development in information retrieval, pp 747–756
36. Liu L, Yang J, Ding Z et al (2023) A sentiment analysis of user attitudes toward information cocoons in short video apps. In: 25th International conference on human-computer interaction, communications in computer and information science, vol 1835. Springer, pp 68–74
37. Liu P, Shivaram K, Culotta A et al (2021) The interaction between political typology and filter bubbles in news recommendation algorithms. In: Proceedings of the web conference, pp 3791–3801
38. Liu Z, Xu W, Zhang W et al (2023) An emotion-based personalized music recommendation framework for emotion improvement. Inf Process Manag 60(3):103256
39. Lu Y, Zhang S, Huang Y et al (2021) Future-aware diverse trends framework for recommendation. In: Proceedings of the web conference, pp 2992–3001
40. Lunardi GM, Machado GM, Maran V et al (2020) A metric for filter bubble measurement in recommender algorithms considering the news domain. Appl Soft Comput 97(Part):106771
41. Luo Z, Huang H, Lian J et al (2023) Cross-links matter for link prediction: rethinking the debiased GNN from a data perspective. In: Advances in neural information processing systems, pp 79594–79612
42. Mansoury M, Abdollahpouri H, Pechenizkiy M et al (2020) Feedback loop and bias amplification in recommender systems. In: The 29th ACM international conference on information and knowledge management, pp 2145–2148
43. Michiels L, Vannieuwenhuyze JTA, Leysen J et al (2023) How should we measure filter bubbles? A regression model and evidence for online news. In: Proceedings of the 17th ACM conference on recommender systems, pp 640–651
44. Min Y, Jiang T, Jin C et al (2019) Endogenetic structure of filter bubble in social networks. R Soc Open Sci 6(11):190868
45. Minici M, Cinus F, Monti C et al (2022) Cascade-based echo chamber detection. In: Proceedings of the 31st ACM international conference on information & knowledge management, pp 1511–1520
46. Morini V, Pollacci L, Rossetti G (2021) Toward a standard approach for echo chamber detection: reddit case study. Appl Sci 11(12):5390
47. Moscati M, Parada-Cabaleiro E, Deldjoo Y et al (2022) Music4all-onion—a large-scale multi-faceted content-centric music recommendation dataset. In: Proceedings of the 31st ACM CIKM, pp 4339–4343
48. Nguyen TT, Hui P, Harper FM et al (2014) Exploring the filter bubble: the effect of using recommender systems on content diversity. In: Proceedings of the 23rd international conference on world wide web, pp 677–686
49. Nian F, Wang Z, Qian Y (2024) Research on homogeneous information propagation in social network. Int J Mod Phys C (IJMPC) 35(06):1–23
50. Nikolov D, Oliveira D, Flammini A et al (2015) Measuring online social bubbles. PeerJ Comput Sci 1:e38
51. Nikolov D, Lalmas M, Flammini A et al (2019) Quantifying biases in online information exposure. J Assoc Inf Sci Technol 70(3):218–229
52. Piao J, Liu J, Zhang F et al (2023) Human-AI adaptive dynamics drives the emergence of information cocoons. Nat Mac Intell 5(11):1214–1224
53. Quattrociocchi W, Scala A, Sunstein CR (2016) Echo chambers on Facebook. Economics of Networks eJournal
54. Raman K, Bennett PN, Collins-Thompson K (2014) Understanding intrinsic diversity in web search: improving whole-session relevance. ACM Trans Inf Syst 32(4):20:1-20:45
55. Santos FP (2023) How to break information cocoons. Nat Mac Intell 5(12):1338–1339
56. Santos FP, Lelkes Y, Levin SA (2021) Link recommendation algorithms and dynamics of polarization in online social networks. Proc Natl Acad Sci USA 118(50):e2102141118
57. Sasahara K, Chen W, Peng H et al (2022) Correction to: Social influence and unfollowing accelerate the emergence of echo chambers. J Comput Soc Sci 5(2):1779
58. Sheng M, Zhang P (2024) "How I form and escape information cocoons": an interview study of users on short video apps. In: Wisdom, well-being, win-win—19th international conference, lecture notes in computer science, vol 14596. Springer, pp 129–138
59. Sinha A, Gleich DF, Ramani K (2016) Deconvolving feedback loops in recommender systems. In: Annual conference on neural information processing systems, pp 3243–3251
60. Spohr D (2017) Fake news and ideological polarization: Filter bubbles and selective exposure on social media. Bus Inf Rev 34(3):150–160
61. Sukiennik N, Gao C, Li N (2024) Uncovering the deep filter bubble: narrow exposure in short-video recommendation. In: Proceedings of the ACM on web conference 2024. ACM, pp 4727–4735. https://doi.org/10.1145/3589334.3648159
62. Sun J, Guo W, Zhang D et al (2020) A framework for recommending accurate and diverse items using Bayesian graph convolutional neural networks. In: The 26th ACM SIGKDD conference on knowledge discovery and data mining, pp 2030–2039
63. Sun J, Song J, Jiang Y et al (2022) Prick the filter bubble: a novel cross domain recommendation model with adaptive diversity regularization. Electron Mark 32(1):101–121
64. Sun W, Khenissi S, Nasraoui O et al (2019) Debiasing the human-recommender system feedback loop in collaborative filtering. In: Companion of the world wide web conference, pp 645–651
65. Tsukuda K, Goto M (2019) Dualdiv: diversifying items and explanation styles in explainable hybrid recommendation. In: Proceedings of the 13th ACM conference on recommender systems, pp 398–402
66. Vrijenhoek S, Bénédict G, Granada MG et al (2022) Radio-rank-aware divergence metrics to measure normative diversity in news recommendations. In: Sixteenth ACM conference on recommender systems, pp 208–219

67. Wang H, Zhang F, Xie X et al (2018) DKN: deep knowledge-aware network for news recommendation. In: Proceedings of the ACM WWW, pp 1835–1844
68. Wang H, Tu S, Gionis A (2024) Sequential diversification with provable guarantees. CoRR abs/2412.10944. https://doi.org/10.48550/ARXIV.2412.10944
69. Wang L, Zhang J, Yang H et al (2025) User behavior simulation with large language model-based agents. ACM Trans Inf Syst. https://doi.org/10.1145/3708985
70. Wang W, Feng F, Nie L et al (2022) User-controllable recommendation against filter bubbles. In: The 45th international ACM SIGIR conference on research and development in information retrieval, pp 1251–1261
71. Wang X, He X, Wang M et al (2019) Neural graph collaborative filtering. In: Proceedings of the 42nd international ACM SIGIR conference on research and development in information retrieval, pp 165–174
72. Wang Z, Shao Y, He J et al (2023) Diversity-aware deep ranking network for recommendation. In: ACM CIKM, pp 2564–2573
73. Wu C, Wu F, An M et al (2019) Neural news recommendation with attentive multi-view learning. In: Proceedings of the twenty-eighth international joint conference on artificial intelligence, pp 3863–3869
74. Wu C, Wu F, Ge S et al (2019) Neural news recommendation with multi-head self-attention. In: Proceedings of the EMNLP. Association for Computational Linguistics, pp 6388–6393
75. Wu F, Qiao Y, Chen J, et al (2020) MIND: a large-scale dataset for news recommendation. In: Proceedings of the 58th annual meeting of the association for computational linguistics, pp 3597–3606
76. Wu L, He X, Wang X et al (2023) A survey on accuracy-oriented neural recommendation: from collaborative filtering to information-rich recommendation. IEEE Trans Knowl Data Eng 35(5):4425–4445
77. Xie R, Liu Q, Liu S et al (2022) Improving accuracy and diversity in matching of recommendation with diversified preference network. IEEE Trans Big Data 8(4):955–967
78. Yang L, Wang S, Tao Y et al (2023) Dgrec: graph neural network for recommendation with diversified embedding generation. In: Proceedings of the sixteenth ACM international conference on web search and data mining, pp 661–669
79. Yang W (2024) Information cocoons on social media: why and how should the government regulate algorithms
80. Ye R, Hou Y, Lei T et al (2021) Dynamic graph construction for improving diversity of recommendation. In: Fifteenth ACM conference on recommender systems, pp 651–655
81. Yuan W, Yang C, Qu L et al (2024) Robust federated contrastive recommender system against model poisoning attack. arXiv preprint arXiv:2403.20107
82. Yuan W, Yuan S, Yang C et al (2024) Manipulating visually aware federated recommender systems and its countermeasures. ACM Trans Inf Syst 42(3):64:1-64:26. https://doi.org/10.1145/3630005
83. Zhang M, Hurley N (2008) Avoiding monotony: improving the diversity of recommendation lists. In: Proceedings of the ACM conference on recommender systems, pp 123–130
84. Zhang X, Cai Y, Zhao M et al (2023) Generation mechanism of "information cocoons'' of network users: an evolutionary game approach. Syst 11(8):414
85. Zheng Y, Gao C, Chen L et al (2021) DGCN: diversified recommendation with graph convolutional networks. In: Proceedings of the web conference, pp 401–412
86. Ziegler C, McNee SM, Konstan JA et al (2005) Improving recommendation lists through topic diversification. In: Proceedings of the 14th international conference on world wide web. ACM, pp 22–32